13. Consider a semicircle with diameter $AB = 2\,\text{cm}$ and centre O. If we take away the semicircle of diameter AO, then we rotate the resulting figure along AB by 360°, we obtain a solid whose volume is:

A. $\frac{25}{3}\pi\ \text{cm}^3$

B. $\frac{7}{6}\pi\ \text{cm}^3$

C. $\frac{5}{6}\pi\ \text{cm}^3$

D. $4\pi\ \text{cm}^3$

E. $\frac{28}{3}\pi\ \text{cm}^3$